

CESSNOCK, Australia

WMO: 957710

Lat: 32.7886S

Lon: 151.3378E

Elev: 62

StdP: 100.59

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.5	-3.8	2.8	14.7	-2.2	3.2	12.4	9.4	15.6	8.4	15.7	0.6	230	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	36.6	20.7	34.1	20.2	32.0	20.0	23.4	30.5	22.6	29.2	21.9	27.8	5.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	16.5	25.1	21.0	15.7	24.3	20.2	15.0	23.8	70.2	30.5	67.1	29.5	64.5	27.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	6.8	-2.6	41.1	1.3	2.3	-3.5	42.7	-4.3	44.0	-5.0	45.3	-5.9	46.9		
			-2.8	24.8	1.3	1.3	-3.8	25.7	-4.5	26.5	-5.3	27.2	-6.2	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	23.6	22.9	21.0	17.3	13.9	11.7	10.8	11.8	14.7	17.4	19.8	22.0
	DBStd	5.21	3.18	2.80	2.57	2.26	2.39	2.25	2.17	2.38	2.69	3.18	3.38	3.07
	HDD10.0	36	0	0	0	0	2	9	16	8	1	0	0	0
	HDD18.3	1023	1	1	5	44	139	200	234	203	113	56	22	5
	CDD10.0	2669	422	362	342	221	122	59	40	64	142	231	294	371
	CDD18.3	615	164	129	89	14	1	0	0	0	4	28	66	118
	CDH23.3	6041	1606	1077	649	162	14	1	1	9	146	459	746	1171
	CDH26.7	2537	783	449	212	32	1	0	0	0	38	169	327	525
Wind	WSAvg	3.0	3.5	3.3	2.9	2.4	2.5	2.7	2.8	3.0	3.1	3.2	3.3	3.4
Precipitation	PrecAvg	746	91	102	88	56	56	49	32	38	40	61	70	67
	PrecMax	1096	251	370	289	246	171	197	127	160	109	188	176	188
	PrecMin	467	2	5	4	2	4	4	0	0	0	1	14	8
	PrecStd	154	59	83	66	58	42	45	29	36	30	47	42	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	38.3	34.6	30.1	25.1	21.0	21.9	24.5	30.4	34.0	37.5	38.4
		MCWB	22.1	21.4	20.9	18.2	15.6	14.8	12.9	13.4	15.2	16.9	19.7	19.9
	2%	DB	36.6	34.5	31.2	27.0	22.9	18.9	19.3	22.1	27.3	30.8	33.5	34.9
		MCWB	21.5	21.4	20.7	18.2	15.4	13.3	11.8	12.7	14.8	17.0	18.9	20.3
	5%	DB	33.9	31.6	28.9	25.0	21.3	17.7	17.7	20.1	24.6	28.3	30.3	32.1
		MCWB	20.9	21.6	20.4	17.5	14.8	12.8	11.6	12.0	14.4	16.9	18.7	19.6
	10%	DB	31.1	29.1	26.8	23.3	19.8	16.7	16.4	18.4	22.2	25.6	27.6	29.5
		MCWB	20.7	20.9	19.9	17.1	14.3	12.6	11.1	11.5	14.1	16.2	18.2	19.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	25.1	23.4	20.5	19.0	16.1	15.8	15.9	18.3	20.5	22.2	23.2
		MCDB	33.5	31.0	28.0	25.0	21.6	18.2	19.0	20.4	23.8	27.5	30.5	31.6
	2%	WB	23.2	23.7	22.3	19.5	17.3	15.0	14.0	14.5	17.0	19.2	21.0	22.1
		MCDB	32.0	29.6	27.3	23.9	19.8	17.3	16.4	18.6	22.1	25.3	28.6	29.4
	5%	WB	22.5	22.8	21.5	18.7	16.2	14.2	13.0	13.5	16.0	18.2	20.0	21.3
		MCDB	29.9	28.2	26.3	23.1	19.4	16.5	15.6	17.7	21.3	24.0	26.8	27.9
	10%	WB	21.8	21.9	20.7	18.0	15.3	13.4	12.0	12.6	15.0	17.2	19.2	20.6
		MCDB	27.8	26.6	25.1	22.1	18.7	15.7	15.2	16.9	20.6	23.3	25.2	26.6
Mean Daily Temperature Range	5% DB	MDBR	12.4	11.2	11.2	12.3	12.6	10.9	12.3	13.9	14.4	14.1	12.9	12.6
		MCDBR	19.2	16.8	15.2	15.5	15.0	12.3	14.8	17.3	19.6	20.2	19.2	18.5
	5% WB	MCWBR	5.7	5.6	6.0	7.4	8.1	7.7	8.6	9.2	8.8	8.2	6.6	6.2
		MCDWR	14.0	12.5	12.2	12.7	11.2	9.9	11.5	13.7	14.5	14.9	14.4	14.2
Clear-Sky Solar Irradiance	taub		0.392	0.384	0.354	0.332	0.312	0.305	0.298	0.305	0.339	0.358	0.393	0.380
		taud	2.442	2.487	2.558	2.592	2.597	2.616	2.616	2.585	2.482	2.457	2.402	2.461
	Ebn at Noon		947	934	928	898	866	848	874	914	928	948	940	961
		Edh at Noon	121	112	98	84	74	69	72	83	103	114	125	120
All-Sky Solar Radiation	RadAvg		6.56	5.74	4.85	3.92	3.12	2.48	2.89	3.83	5.02	5.75	6.15	6.64
	RadStd		0.44	0.51	0.43	0.33	0.24	0.20	0.23	0.25	0.28	0.52	0.58	0.57

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.62	N/A	-1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.45	N/A	N/A	+0.85	N/A	N/A	-2	-85	+189	+102

Nomenclature: See separate page